

Auditing Disocclusion Reliability Evidence: Diagnostic Separability Does Not Imply Observable Routing

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Abstract

We audit whether a slow generative teacher’s usefulness in single-view 3D reconstruction can be predicted from inference-time evidence. The previously reported frame result—ROC-AUC 0.947, trapezoidal PR-AUC 0.961, and +1.604 dB at 41% calls saved—uses ground-truth-derived PSNR quality probes. It is a diagnostic, not deployment performance. On the same 2,898 frames and 74 scene-grouped folds, camera translation and rotation yield accuracy 0.612, ROC-AUC 0.650, trapezoidal PR-AUC 0.663, and +0.715 dB, retain a −23.094 dB worst case, and make 2,028 calls. E-223 combines RGB disagreement with `visibility.npy`, which VGGT derives from the complete target clip; it is therefore an oracle-visibility diagnostic. On a selected population of 910 frames from 21 high-gain scenes (96.5% positive), it reaches ROC-AUC 0.426, average precision 0.929, and calls every frame at threshold 0.5. Thus the release supports diagnostic separability but no operational routing claim.

1 Introduction

A generative prior can improve disoccluded regions in feed-forward novel-view reconstruction [1] but can also cause severe losses. Reliability prediction would be useful only if its inputs exist before GT and teacher evaluation. We revisit the evidence under this requirement and separate four tiers: (i) oracle true gain, (ii) GT-quality-probe diagnostic, (iii) camera-only observable evidence, and (iv) RGB disagreement + oracle-visibility diagnostic evidence.

Contributions.

1. We correct the provenance and semantics of the released frame features, including the historical `vis_gap` field.
2. We reproduce the full-population camera-only grouped-CV baseline and contrast it with the quality-probe diagnostic upper bound.

3. We report the RGB-disagreement result as a selected-population negative result and add executable feature-whitelist and claim-integrity checks.

2 Evidence Audit

Labels and oracle. For frame i , gain is the teacher’s invisible-region PSNR minus the baseline’s invisible-region PSNR, both evaluated against GT. The binary label is one when gain exceeds 0.1 dB. Oracle decisions use true gain and are upper bounds.

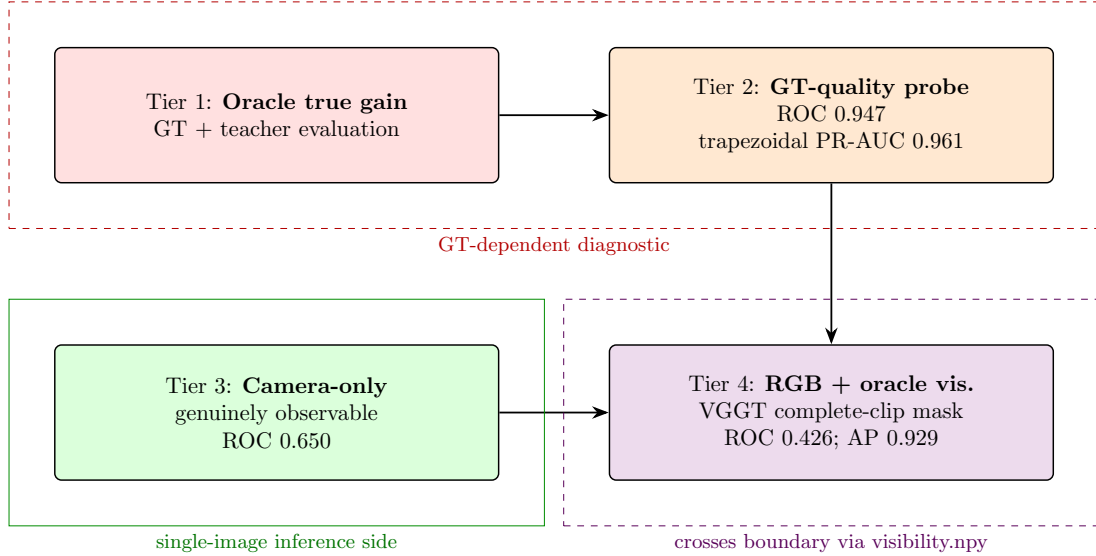
Corrected feature semantics. The frame JSON is preserved, but its column names require care. `f3d_vis` and `f3d_inv` are PSNR quality probes computed against GT. Historical `vis_gap` equals `f3d_vis - base_inv`; it is not an expert-visible minus baseline-visible difference. Interactions built from these columns remain GT-derived. Camera translation and rotation are observable.

Evaluation. All frame experiments use grouped leave-one-scene-out cross-validation: every frame from the held-out scene is excluded from fitting. The camera-only and quality-probe frame experiments share 2,898 frames and 74 scenes. The RGB experiment instead uses 910 frames from 21 selected high-gain scenes, so it is reported separately rather than as a controlled comparison.

3 Results

3.1 Quality probes reveal diagnostic separability

E-145b reaches ROC-AUC 0.947 and trapezoidal PR-AUC 0.961. Its threshold-0.5 policy obtains +1.604 dB and skips 41% of calls. These numbers show that frame outcomes have strong structure when GT-derived reconstruction quality is supplied. They do not measure inference-time prediction. The old frontier and routing



Audit conclusion: only camera translation and rotation are genuinely observable here.
E-223 is selected, not population-aligned, and uses visibility derived by VGGT from the complete target clip. No deployment claim.

Figure 1: Four evidence tiers. Oracle true gain and GT-quality probes are GT-dependent diagnostics. E-223 also crosses the boundary because VGGT derives its visibility mask from the complete target clip. Only E-224 camera translation and rotation are genuinely observable. No deployment claim is made.

Table 1: Four evidence tiers. The first three rows share the 2,898-frame population; E-223 uses 910 frames from 21 selected scenes. Precision-recall metrics are labeled per row because E-223 reports average precision whereas E-145b/E-224 use trapezoidal integration.

Tier	Population	Predictor inputs	Acc.	ROC	Precision-recall	Mean / policy
Oracle true gain	2,898 / 74 scenes	True gain	—	—	—	+1.70 dB upper bound
GT-quality-probe diagnostic	2,898 / 74 scenes	GT PSNR probes + camera	0.861	0.947	Trap. PR-AUC 0.961	+1.604 dB; 41% saved
Camera-only observable	2,898 / 74 scenes	Translation + rotation	0.612	0.650	Trap. PR-AUC 0.663	+0.715 dB; 2,028 calls
RGB + oracle-vis. diagnostic	910 / 21 selected	RGB + VGGT visibility + camera	0.965	0.426	Average precision 0.929	all 910 called

strip are kept only as diagnostic views of this upper bound.

3.2 Observable camera evidence is insufficient

E-224 admits exactly two predictors: target-camera translation and rotation relative to the source. On all 2,898 frames it reaches accuracy 0.6118, ROC-AUC 0.6496, and trapezoidal PR-AUC 0.6631. At threshold 0.5 it calls 2,028 frames, saves 870 (30.0%), and obtains +0.7152 dB mean gain. Its −23.0944 dB worst case is the same severe loss exposed by always calling, so the result does not establish a safe policy.

3.3 RGB disagreement + oracle-visibility diagnostic

E-223 uses baseline/Flash3D disagreement, image gradients, camera motion, and `visibility.npy`. VGGT derives this mask from the complete target clip, so it crosses the single-image inference boundary. Its 21 scenes are selected by complete-asset availability in the high-gain `teacher.npy` directory, not population-aligned with E-145b/E-224; exact IDs and counts are released in a manifest. Of 910 frames, 878 (96.5%) are positive. ROC-AUC is 0.4255, average precision is 0.9290, and the threshold-0.5 model calls all frames. The high accuracy and average precision reflect prevalence, not useful ranking.

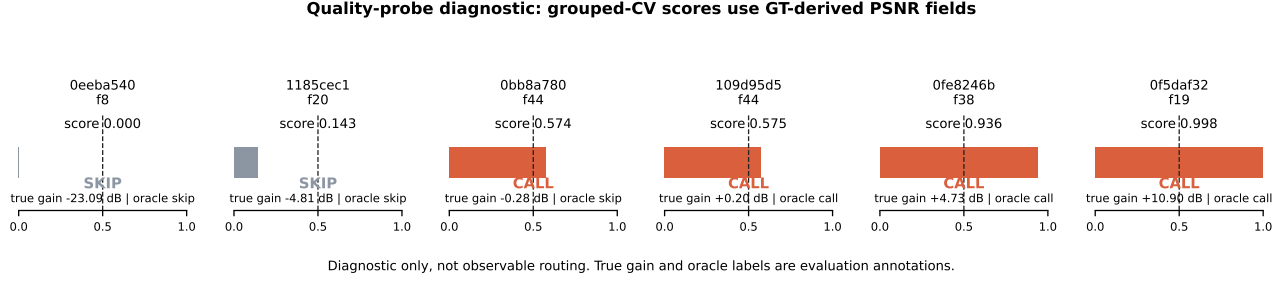


Figure 2: Quality-probe diagnostic strip from the released E-145b JSON inputs. The grouped-CV scores depend on GT-derived PSNR probes. True gain and labels are evaluation annotations. This illustrates diagnostic separability, not an observable or operational routing process.

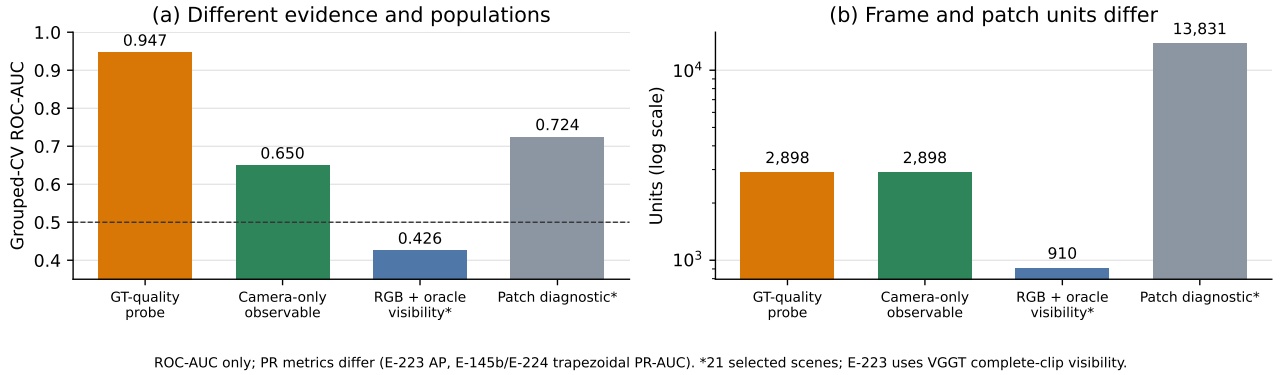


Figure 3: Diagnostic boundary, not a controlled leaderboard. Frame quality-probe results use 2,898 frames from 74 scenes and report PSNR-based policy gain; patch results use 13,831 patches from 21 selected high-gain scenes and patch-MSE-based gain. Populations, units, prevalence, inputs, and gain metrics differ.

4 Discussion

The audit supports a narrower scientific result than the prior routing framing. Teacher usefulness is highly separable at frame level with GT-derived quality probes, which provides a useful target for future uncertainty estimators. However, camera motion gives only modest full-population discrimination, while E-223 fails to rank a heavily imbalanced selected set even with oracle visibility. A valid operational claim requires stronger observable signals evaluated on the same unselected population, followed by end-to-end latency and safety testing.

5 Limitations

E-223 and the patch study use selected high-gain scenes and cannot be directly compared with the 74-scene frame population. E-223’s rendered and VGGT visibility assets are external, so only its aggregate and manifest are released. E-224 is deliberately simple and does not rule out better observable uncertainty features. The audit changes interpretation, not released

measurements.

6 Conclusion

GT-derived quality probes reveal strong frame-level diagnostic separability, but this is not evidence of deployable routing. The full-population camera-only baseline is substantially weaker, and the selected-population RGB + oracle-visibility diagnostic is negative and limited. The evidence tiers and executable guards make that boundary explicit.

References

- [1] Stanislaw Szymanowicz, Eldar Insafutdinov, Chuanxia Zheng, Dylan Campbell, Joao Henriques, Christian Rupprecht, and Andrea Vedaldi. Flash3d: Feed-forward generalisable 3d scene reconstruction from a single image. In *arXiv preprint arXiv:2406.04343*, 2024.